

Implicit Modeling and Solving of an (MI)LP with JuMP

Writing an implicit model (1/5)

Example (cont'd):

$$\left[\begin{array}{llll} \max f(x) = & x_1 & + & 3x_2 & (0) \\ \text{s.t} & x_1 & + & x_2 & \leq 14 \quad (1) \\ & -2x_1 & + & 3x_2 & \leq 12 \quad (2) \\ & 2x_1 & - & x_2 & \leq 12 \quad (3) \\ & x_1 & , & x_2 & \geq 0 \quad (4) \end{array} \right]$$

↓

$$c = (1, 3), \quad d = (14, 12, 12), \quad T = \begin{pmatrix} 1 & 1 \\ -2 & 3 \\ 2 & -1 \end{pmatrix}$$

$$\left[\begin{array}{llll} \max f(x) = & \sum_{j=1}^2 c_j x_j & & (0) \\ \text{s.t} & \sum_{j=1}^2 t_{ij} x_j \leq d_i & i = 1, 3 & (1 - 3) \\ & x_j \geq 0 & j = 1, 2 & (4) \end{array} \right]$$



Example (cont'd):

$$c = (1, 3), \quad d = (14, 12, 12), \quad T = \begin{pmatrix} 1 & 1 \\ -2 & 3 \\ 2 & -1 \end{pmatrix}$$

```
julia> c = [1, 3]
julia> d = [14, 12, 12]
julia> T = [1 1 ; -2 3; 2 -1]
julia> n,m = size(T)
```



Example (cont'd):

$$\left[\begin{array}{ll} \max z(x) = & \sum_{j=1}^m c_j x_j & (0) \\ \text{s.t} & \sum_{j=1}^m t_{ij} x_j \leq d_i & i = 1, n \quad (1-3) \\ & x_j \geq 0 & j = 1, m \quad (4) \end{array} \right]$$

```
julia> using JuMP, HiGHS
julia> md = Model(HiGHS.Optimizer)
```

```
julia> @variable(md, x[1:m] ≥ 0)
```

```
julia> @objective(md, Max, sum(c[j]*x[j] for j=1:m))
```

```
julia> @constraint(md, cst[i=1:n], sum(T[i,j]*x[j]
                                     for j=1:m) ≤ d[i])
```



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```

Structured variables in JuMP

- ▶ one-based integer ranges...

```
julia> @variable(md, x[1:4] ≥ 0)
```

```
julia> @variable(md, x[1:2, 1:2] ≥ 0, Int)
```

- ▶ ...but not necessarily...

```
julia> @variable(md, x[-4:4] ≥ 0)
```

- ▶ ...nor necessarily numerical

```
julia> @variable(md, x[:, :tram, :train]), Bin)
```

More on solutions

Primal solution values for a scalar variable:

```
value(variable)
```

```
julia> value(x1)
```

Primal solution values for a structured variable:

```
value(variable[index])
```

```
julia> value(x[2])
```

```
value.(variable)
```

```
julia> value.(x)
```

More on solutions

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```
value(variable)
```

```
julia> value(x1)
```

Primal solution values for a structured variable:

```
value(variable [index])
```

```
julia> value(x[2])
```

```
value. (variable)
```

```
julia> value. (x)
```